11/10 K3 reading group. Hajime Nakahashi

"Cones and Symmetries of K3 Surfaces"

§1 The Geometry of Cones on K3 surfaces.
§2 Cones in Higher Dimensions.
§3 Cone Conjecture and Beyond.

§1

1-1 The Cone of curves and Its Dual

X: proj K3 surf

Def

The **Effective Cone** is

$$NE(X) = \left\{ \sum_{i=1}^{r} a_i [C_i] \;\middle|\; a_i \geq 0, C_i \subset X \text{ irred curves} \right\}$$

Its closure is called the **Mori Cone**: $\overline{NE}(X)$

Thm (Huy Ch 8. Thm 3.2)

$$Nef(X) = NE(X)^* = \overline{NE}(X)^*$$

more precisely; $Nef(X) = \{\alpha \mid (\alpha.\beta) \geq 0 \text{ for all } \beta \in NE(X)\}$

$\overline{NE}(X) = \{\beta \mid (\alpha.\beta) \geq 0 \text{ for all } \alpha \in Nef(X)\}$

Q: What determines the shape of the Mori cone?

1-2 Two Sources of the Boundary

C: irred curve

adjunction gives $C^2 = 2Pa(C) - 2$

In particular

$$C^2 < 0 \iff C \simeq \mathbb{P}^1, \; C^2 = -2$$

On the other hand, Riemann-Roch gives

$$\chi(\mathcal{O}_X(D)) = 2 + \frac{D^2}{2} > 0 \quad (\text{for } D \in \mathcal{C}_X)$$

$\Rightarrow D$: eff (i.e $\overline{\mathcal{C}_X} \subsetneq \overline{NE}(X)$)

$\mathcal{C}_X$: positive cone containing the ample classes

These observations lead to

$$\boxed{\overline{NE}(X) = \overline{\mathcal{C}_X} + \overline{\sum_{\substack{C \subset X \\ C^2 = -2}} \mathbb{R}_{\geq 0}[C]}}$$

Two sources of boundary rays:

$\underbrace{D^2 = 0}$ and $\underbrace{C^2 = -2}$

Isotropic directions — Rational curves

1-3 An elliptic K3 surface

Consider an elliptic K3 surface

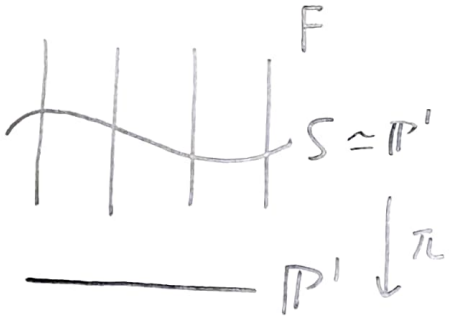
$$\pi : X \longrightarrow \mathbb{P}^1 \quad (\text{very general})$$

with a section S and fiber class F.

Assume $\boxed{NS(X) = \mathbb{Z}S \oplus \mathbb{Z}F}$

with intersection matrix

$$\begin{pmatrix} S^2 & S.F \\ S.F & F^2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix}$$



(1) The Mori cone

Let $C \subset X$ be an irred curve w/ $C \neq S$

Write $[C] = a[S] + b[F]$ $\quad a, b \in \mathbb{Z}$

Then $a = C.F \geq 0$, $\quad b - 2a = C.S \geq 0$

Hence $a, b \geq 0$

$\Rightarrow \boxed{\overline{NE}(X) = \text{Cone}(S, F)}$

(2) The nef cone

By duality, for $D = xS + yF$

$$D \in Nef(X) \Leftrightarrow \begin{cases} D \cdot F = x \geq 0 \\ D \cdot S = y - 2x \geq 0 \end{cases}$$

since $D = x(S+2F) + (y-2x)F$

we obtain

$$\boxed{Nef(X) = Cone(F, S+2F)}$$

(3) The positive cone

$$D^2 = 2x(y-x)$$

Choosing the component containing ample classes,

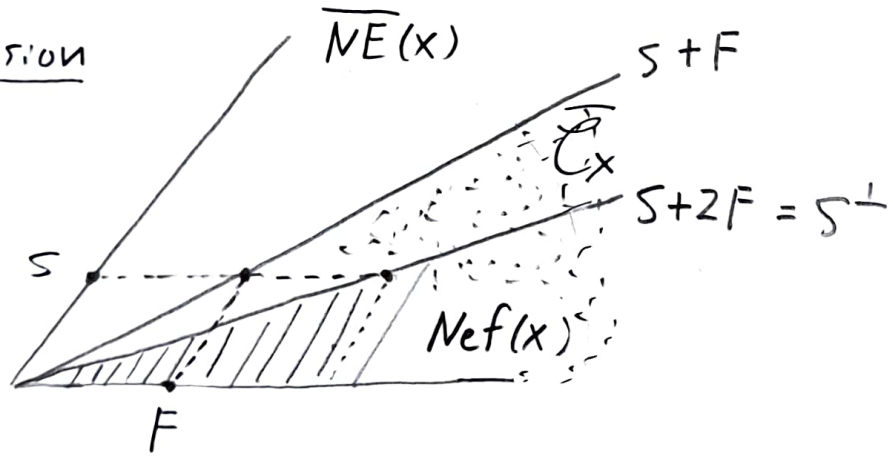
$$\mathcal{C}_X = \{xS + yF \mid x>0, y>x\}$$

Since $D = x(S+F) + (y-x)F$

we get

$$\boxed{\overline{\mathcal{C}_X} = Cone(F, S+F)}$$

Conclusion



Note The two boundaries of the nef cone have different geometric meanings

F : an isotropic class defining the elliptic fibration

$S+2F$: orthogonal to the rational curve S

Q: How much freedom do we have in combining these two types of boundaries?

⟶ Kovács' classification.

1-4 Possible shapes of Cones of K3 Surfaces.

① $\rho = 2$

Two questions:

1. Does $NS(X)$ contain a nonzero isotropic class?
2. Does X contain a (-2)-curve?

For $\rho(X) = 2$, the two isotropic boundary rays are either both rational or both irrational. (Since the intersection form is a binary quadratic form over the rationals.)

⇝ Four cases (and all four possibilities actually occur)

	$\nexists_{(-2)}$	$\exists_{(-2)}$
$\nexists$ isotropic	Ex 1 	Ex 4 
$\exists$ isotropic	Ex 2 	Ex 3 

② $\rho \geq 3$

a hyperplane section of the positive cone is a ball.

	$\nexists_{(-2)}$	$\exists_{(-2)}$
$\nexists$ isotropic	Ex 5 	Ex 7 
$\exists$ isotropic	Ex 6 	? Can two types coexist? → No!


allow 

1-5 Kovács' Theorem

Thm (Kovács)

X : proj K3 surf w/ $\rho(X) \geq 3$

Then
$$\overline{NE}(X) = \begin{cases} \overline{C}_X & \text{if } \nexists (-2)\text{-curves} \\ \overline{\sum_{C^2=-2} \mathbb{R}_{\geq 0}[C]} & \text{otherwise.} \end{cases}$$

In particular, the Mori cone is either entirely round or has no round part.

Recall
$$\overline{NE}(X) = \overline{C}_X + \overline{\sum \mathbb{R}_{\geq 0}[C]}$$

Kovács implies: If $\rho \geq 0$ and $\exists (-2)$-curve
$$\Rightarrow \overline{C}_X \subseteq \overline{\sum \mathbb{R}_{\geq}[C]}$$

Idea of proof

key: Approximation of extremal rays ✧

Consequences

- $\rho \geq 3$: No mixed circular and rational curve boundaries

(
- Case (v) : $\rho(X) \leq 4$ (by Meyer's thm)
- Case (vi) : $\rho(X) \leq 11$ (by Huybrechts Ch 14, Cor 3.8)
)

- $\rho \geq 12$: Every proj K3 contains (-2) curve.
  $\rho \geq 12 \Rightarrow NS(X) \simeq U \oplus N$

1. The arithmetic of the Néron-Severi lattice imposes surprisingly strong restrictions on the geometry of the Mori cone.

§2 Cones in Higher Dimensions

Q: What Do the Chambers Represent?

2-1 Revisiting the K3 Picture

Recall

- chamber decomp $= C_X \setminus \bigcup_{\delta^2=-2} \delta^\perp$
- Weyl group $W_X \curvearrowright \{\text{chambers}\}$
- $\mathrm{Amp}(X)$ is a fundamental chamber.

Q: What do the other chamber represent?

$\longrightarrow$ the same K3 surface with different marking

2-2 A special feature of K3 surfaces

$$\boxed{X \dashrightarrow X' \Rightarrow X \cong X'}$$

Birational K3 surfaces are isomorphic.

2-3 The Higher Dimensional Picture

K3 $\overset{\text{high-dim generalization}}{\rightsquigarrow}$ Irreducible Holomorphic Symplectic manifolds (a.k.a cpt HyperKähler)

X: IHS of dim $2n$.

the intersection form is replaced by the Beauville-Bogomolov-Fujiki form:

$$\boxed{\int_X \alpha^{2n} = C_X q_X(\alpha)^n} \quad \forall \alpha \in H^2(X, \mathbb{Z}),\ C_X: \text{Fujiki const}$$

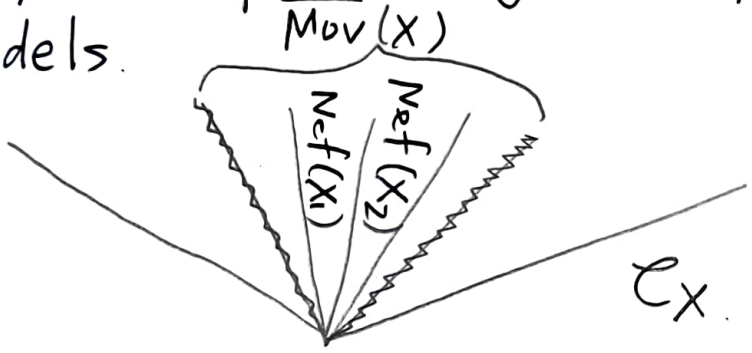
Hence we can again define

$$\mathcal{C}_X = \{\alpha \in NS(X)_{\mathbb{R}} \mid q_X(\alpha) > 0\}^+$$

But now

$$\boxed{Amp(X) \subset Mov(X) \subset \mathcal{C}_X}$$

Inside the movable cone, different ample chambers may correspond to genuinely different birational models.



2-4 Two types of walls

	⟿ Divisorial wall	— Flopping wall
Cont.	Divisor	Codim ≥ 2
Position	Boundary of Mov.	Inside Mov.
Wall cross	Reflection	Flop
e.g	$S^{[2]} \to S^{(2)}$	Mukai flop

Monodromy Birationally Minimal class

see Amerik - Verbitsky

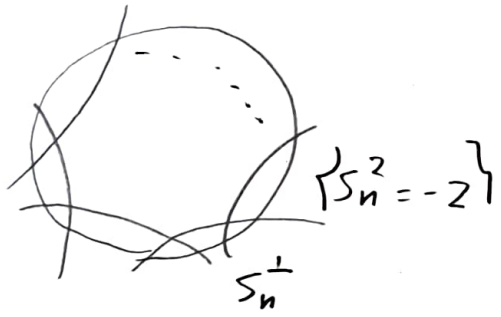
§3 Cone Conjecture

3-1 From Kovács to Sterk

Kovács : which curves determine the cone?

New question : what if there are infinitely many such curves?

e.g (vii) elliptic K3 surf w/ infinitely many section.
(Mordell-Weil rank 1 $MW(X) \simeq \mathbb{Z}$)



3-2 Sterk's theorem

= Although the nef cone may have infinitely many walls, its structure becomes finite up to automorphism.

⚠ W_X : acts on the positive cone

‖ $\mathrm{Aut}(X)$: preserves the nef cone.

We define the effective Nef cone ;

$$Nef^e(X) = Conv_{\mathbb{R}}(Nef(X) \cap NS(X)) = \sum_{\substack{L \in NS(X) \\ L : nef}} \mathbb{R}_{\geq 0}[L].$$

e.g for (i)

$Nef^e(X) = \mathcal{C}_X \cup \{0\}$, $Nef(X) = \overline{\mathcal{C}_X}$

↗ contains irrational isotropic ray.

Thm (Sterk)

X : proj K3 surface

$\exists\ \Pi$: rational polyhedral fundamental domain for the action

$$Aut(X) \curvearrowright Nef^e(X)$$

In particular $Nef^e(X) = \bigcup_{g \in Aut(X)} g^* \Pi$

Idea of proof $W_X \lhd Aut_s(X) < O^+(NS(X))$ (finite index)

3-3 Cone conjecture

higherdim ⇝ Morrison - Kawamata Cone Conjecture

X : CY (K-trivial)

$Mov(X) = Cone_{\mathbb{R}} \{[D] \mid \operatorname{codim}_X Bs|D| \geq 2\}$

We expect rational polyhedral fundamental domains for $Aut(X) \curvearrowright Nef^e(X)$

$Bir(X) \curvearrowright \overline{Mov}^e(X)$.

3-4 Current Status

K3	✓ (Sterk)
IHS	Major general result (Markman, Amerik-Verbitsky)
Sing Symp	"Lehn - Mongardi - Pacienza"
strict CY	open

3-5 Final Remark

From Algebraic to Complex K3 surfaces

$$NS(X)_{\mathbb{R}} \subset H^{1,1}(X,\mathbb{R})$$

C_X^{an} : positive cone in $H^{1,1}(X,\mathbb{R})$

$$K_X = \{\alpha \in C_X^{an} \mid (\alpha, C) > 0 \text{ for every } C \simeq \mathbb{P}^1\}$$

- Weyl chamber $\rightsquigarrow$ Pwp 5.5
- Kovács, Sterk $\rightsquigarrow$ X (see §5.4)